TEXORA · LECTURE SHEET

Weft Knitting Calculations

Every production question in weft knitting reduces to five quantities — needle count, courses per minute, yarn length per course, yarn count, and relaxed CPC/WPC — combined by unit discipline. This sheet derives each formula once, then works the standard problems: production in kg/h and m/h, fabric weight, width, yarn requirement and efficiency.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 12 · LEVEL 3 TERM 1

1. The five base quantities

$$N = \pi \times D \times E \quad \cdot \quad \text{Courses/min} = F \times \text{rpm (circular)} \quad \cdot \quad \text{Yarn per course} = (\text{needles knitting}) \times l$$

- **N** — needles in the bed: D in inches, E needles/inch. Rib/interlock: N per bed; count how many needles knit per course (Topic 5: both beds; Topic 6: half of each bed per feeder).
- **l** — stitch length, mm per loop.
- **tex** — yarn linear density, g per 1000 m. Conversions: tex = 590.5 ÷ Ne (cotton count); denier = 9 × tex.
- **CPC, WPC** — always the *relaxed* values for length/weight/width of delivered fabric.
- **Efficiency η** — multiply theoretical output by η (decimal) at the end, once.

2. Production in kg/h — derivation and master example

Yarn length knitted per minute = (courses/min) × (yarn per course). Convert to km/h, multiply by tex (g/km) to get g/h, divide by 1000 for kg/h:

$$\text{kg/h} = F \times \text{rpm} \times 60 \times N_{knit} \times l(\text{mm}) \times \text{tex} \div 10^9 \times \eta$$

MASTER EXAMPLE — SINGLE JERSEY (ALL NUMBERS RE-VERIFIED)

D = 30 in, E24, F = 72, rpm = 25, l = 5.0 mm, tex = 30, η = 0.85.

N = π × 30 × 24 = 2262. Courses/min = 72 × 25 = 1800.

Yarn/min = 1800 × 2262 × 5.0 mm = 20,358,000 mm = 20,358 m. Per hour: 1,221,480 m = 1221.48 km.

Mass = 1221.48 × 30 = 36,644 g/h = 36.64 kg/h. × 0.85 = **31.1 kg/h**. ✓ (Same machine as Topic 4.)

3. Production in m/h, width and GSM

$$\text{Length (m/h)} = \text{Courses/min} \times 60 \div \text{CPC(per cm)} \div 100 \times \eta \quad \cdot \quad \text{Open width (cm)} = N \div \text{WPC} \quad \cdot \quad \text{GSM} = \text{CPC} \times \text{WPC} \times l \times \text{tex} \div 100$$

- **Length:** courses/hour divided by courses per cm gives cm/h. Interlock: use interlock courses/min (= F × rpm ÷ 2).
- **Width:** a tube opened once lays all N wales across the width. For a garment tube used uncut, the lay-flat width is N ÷ WPC ÷ 2.
- **GSM derivation:** loops/m² = CPC × WPC × 10,000 (per cm² → per m²). Yarn per m² = loops × l mm = CPC × WPC × l × 10 m… collecting: mass = CPC × WPC × l(mm) × tex(g/km) ÷ 100 → g/m². Check with Topic 1 numbers: CPC 15, WPC 12, l 4.2, tex 19.7 → 15 × 12 × 4.2 × 19.7 ÷ 100 = **148.9 ≈ 149 g/m² ✓**.
- **Area check:** kg/h ÷ (m/h × width in m) must return the GSM in kg/m² — use it to catch unit errors.

WORKED EXAMPLE — FULL SET FOR THE MASTER MACHINE

Relaxed values measured on this fabric: CPC = 11/cm, WPC = 9/cm.
Length = 1800 × 60 ÷ 11 ÷ 100 × 0.85 = 108,000 ÷ 1100 × 0.85 = 98.2 × 0.85 = **83.5 m/h**.
Open width = 2262 ÷ 9 = **251.3 cm**; lay-flat tube width = 125.7 cm.
GSM = 11 × 9 × 5.0 × 30 ÷ 100 = **148.5 g/m²**.
Area cross-check: 31.1 kg/h ÷ (83.5 m/h × 2.513 m) = 31,100 ÷ 209.8 = 148.2 ≈ 148.5 g/m² ✓ (difference is rounding).
Consistency rule: never mix a stitch length from one fabric with CPC/WPC from another. The three quantities relax together — measure all three on the same sample. A CPC of 20/cm, for instance, belongs to a much shorter stitch length and finer yarn, and pairing it with l = 5.0 mm would fabricate a false GSM.

4. Rib and interlock calculations

MACHINE	NEEDLES KNITTING PER COURSE	COURSES COUNTED FOR LENGTH
Single jersey	N	F × rpm
Rib (per rib course)	2N (both beds)	F × rpm rib courses

Interlock (per feeder)	N (½ of each bed × 2 beds)	F × rpm ÷ 2 interlock courses

Rib check (Topic 5): kg/h theoretical = 960 courses/min × 2 × 1696 × 5.5 × 29.5 ÷ 10^9 × 60 = 31.71 kg/h ✓. Interlock check (Topic 6): 960 × 2262 × 4.6 × 19.7 ÷ 10^9 × 60 = 11.81 kg/h ✓. The master formula reproduces both earlier sheets — always re-derive, never memorise a special case.

5. Yarn requirement and order planning

Yarn for an order (kg) = fabric area (m²) × GSM (kg/m²) × (1 + waste allowance)

Example: 5000 m of 1.60 m open-width jersey at 150 GSM, 3 % knitting waste: area = 5000 × 1.60 = 8000 m²; yarn in fabric = 8000 × 0.150 = 1200 kg; with waste = 1200 × 1.03 = **1236 kg** to order. Machine time at 31.1 kg/h (Topic 4 rate, comparable fabric): 1200 ÷ 31.1 = **38.6 machine-hours** of productive knitting — schedule from this, adding doffing and changeover outside the efficiency already applied.

6. Efficiency and speed factor

Efficiency η = actual output ÷ theoretical output at the same settings over the same period (weigh a roll and divide by its running time). Speed factor = D(m) × rpm compares mechanical loading across sizes: master machine = 0.762 × 25 = **19.1**. Raising rpm raises kg/h linearly but raises yarn-break and fault rates faster than linearly past the machine's comfortable factor — quote both numbers in any speed proposal.

Glossary

Theoretical production — output at η = 100 %. **Waste allowance** — extra yarn for press-offs, changeovers and faulty lengths. **Consistency rule** — l, CPC and WPC in one calculation must come from the same relaxed sample.

Key points

• One master formula serves jersey, rib and interlock — only "needles knitting per course" and "courses for length" change.
• GSM = CPC × WPC × l × tex / 100; width = N / WPC; length from courses and CPC.
• Cross-check kg/h ÷ (m/h × width) = GSM; apply efficiency once, at the end.

Practice

1. Jersey: D 28 in, E24, F 84, rpm 28, l 4.6 mm, tex 19.7, η 0.88. Find kg/h. (N = π × 28 × 24 = 2111.)
2. For Q1's machine, relaxed CPC = 16/cm: find m/h.
3. Find GSM for CPC 14, WPC 11, l 4.8 mm, tex 24.6.
4. An order needs 3000 m² at 180 GSM with 4 % waste. How much yarn?
5. A rib machine's kg/h is correct but its m/h was calculated using F × rpm as interlock courses. State the error's direction and size factor for an interlock conversion of the same numbers.

Answer key

1. Courses/min = 84 × 28 = 2352. Yarn/min = 2352 × 2111 × 4.6 mm = 22,839,331 mm ≈ 22,841 m (using the unrounded N) → per hour ≈ 1,370,460 m = 1370.5 km. Mass = 1370.5 × 19.7 = 27,000 g ≈ 27.0 kg/h; × 0.88 = **23.8 kg/h**.
2. Length = 2352 × 60 ÷ 16 ÷ 100 × 0.88 = 141,120 ÷ 1600 × 0.88 = 88.2 × 0.88 = **77.6 m/h**.
3. GSM = 14 × 11 × 4.8 × 24.6 ÷ 100 = **181.8 g/m²**.
4. Yarn = 3000 × 0.180 × 1.04 = **561.6 kg**.
5. Interlock courses are F × rpm ÷ 2, so using F × rpm doubles the course count: the m/h is overstated by a factor of **2**.

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